# GIOIA ZHENG

M.Sc. Computer Science · Reinforcement Learning · Computer Vision · Reproducible ML
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Computer science master's student with hands-on projects in reinforcement learning, handwriting recognition and NLP evaluation. Currently starting an IAS robot-learning project, with experience in Python, PyTorch, testing and reproducible experiments.

## EDUCATION

**TU Darmstadt**
M.Sc. Computer Science · Enrolled, Winter Semester 2026/27
Selected for Robot Learning: Integrated Project - Part 1 (20-00-0753-pj), WiSe 2026/27
Project (ongoing): Learning Amphibious Legged Locomotion
Intelligent Autonomous Systems (IAS) Lab · Course instructor: Prof. Jan Peters
Project supervisors: Henrik Metternich, Davide Molinaroli, Sucheth Shenoy

**Sapienza University of Rome**
B.Sc. Applied Computer Science and Artificial Intelligence · Expected Dec 2026

## RESEARCH AND ENGINEERING EXPERIENCE

**Undergraduate Thesis Research — RAG Pipeline Evaluation**
Sapienza University of Rome · Aug 2026–Present
Supervisor: Prof. Marco Raoul Marini · Co-tutor: Valerio Venanzi

- Evaluated retrieve-rerank-generate pipelines across MS MARCO, TREC-DL, SciFact, and NFCorpus.
- Found 72/323 NFCorpus queries with no relevant document in the top 100, including 48 still unresolved at depth 1,000; distinguished depth-recoverable misses from persistent retrieval failures.
- Compared ranking improvements with fixed candidate recall to identify when reranking cannot recover missing evidence.

**NLP Algorithm Engineer Intern**
Microsoft · Remote · Feb–May 2026

- Implemented BM25 retrieval and cross-encoder reranking for an MS MARCO question-answering pipeline in Python.
- Extended evaluation to TREC-DL 2019/2020 and built reproducible experiment, metric-validation, and query-level analysis workflows.

## SELECTED MACHINE LEARNING PROJECTS

**Q-learning Exploitability** Python · Reinforcement Learning · Self-Play

- Implemented tabular self-play Q-learning and evaluated policies against exact best responses to expose failures hidden by sampled win rates.
- Compared opponent mixtures, adversarial Bellman backups and symmetry-aware state representations under matched training budgets.

**Handwritten OCR System** PyTorch · OpenCV · CNN–RNN–CTC

- Developed an experimental handwriting-recognition pipeline with image preprocessing, convolutional features, bidirectional LSTMs and CTC decoding.
- Implemented image augmentation and CER/WER evaluation; documented dataset configuration and reproducibility checks.

## ADDITIONAL EXPERIENCE

**NoDiscorSi** Social Media Support

- Helped manage Instagram and Facebook accounts.

## OPEN SOURCE CONTRIBUTIONS

- Merged MTEB contributions: symmetric STS pair counting (#4958), GermanGovService v2 (#5323), and evaluation-request workflow documentation (#4861).
- Merged fixes for cached M-BEIR instruction lookup in Pyserini (#2655) and model language-scope display in the MTEB leaderboard frontend (#31).

## TECHNICAL SKILLS

Programming: Python, SQL, Bash, Git; working knowledge of Go, JavaScript, and Java
ML and Vision: PyTorch, OpenCV, CNNs, LSTMs, CTC, tabular Q-learning, Transformers, scikit-learn
Evaluation: Policy robustness, CER/WER, MRR, nDCG, paired bootstrap, ablations and failure analysis
Engineering: Linux, Docker, GitHub Actions, pytest, SQLite, REST APIs, reproducible configurations and CI
Languages: Chinese (native), Italian (bilingual), English (professional), German (beginner)